Elements of AIML
Unit 03 – Lecture 03 Notes
Dimensionality Reduction (PCA, LDA, ICA)

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1 Lecture Overview

This lecture introduces **dimensionality reduction** and focuses on PCA, with conceptual overviews of LDA and ICA. Dimensionality reduction is important for:

- faster training/inference,
- reducing noise and redundancy,
- visualization of high-dimensional datasets.

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.

2 How to use these notes (self-study)

First, build intuition: PCA finds directions of maximum variance and projects data onto them. Then study the step-by-step pipeline (centering, covariance, eigenvectors, projection) and work through the practice questions. When using PCA in code, remember the evaluation rule: fit PCA on training data only to avoid leakage.

3 Learning Outcomes

- Explain the curse of dimensionality at an intuitive level.
- Describe PCA and interpret principal components and explained variance.
- Outline PCA steps and apply PCA using Python.
- Distinguish PCA vs LDA vs ICA conceptually.

4 Keywords

- **Dimensionality:** number of features (d).
- **Principal component:** direction capturing maximum variance, orthogonal to previous.
- **Explained variance:** variance captured by a component (often ratio).
- **Projection:** mapping data to a lower-dimensional space.
- **Covariance matrix:** captures relationships between features.

5 Abbreviations and symbols

- n: number of samples (rows), d: original number of features, k: reduced number of components.
- $X \in \mathbb{R}^{n \times d}$: data matrix; X_c: mean-centered data.
- C: covariance matrix; λ_i: eigenvalues; v_i: eigenvectors.
- V_k: matrix of top-k eigenvectors; $Z = X_c V_k$: projected data.
- **EVR**: explained variance ratio $r_i = \lambda_i / \sum_j \lambda_j$.

6 Why dimensionality reduction?

6.1 Curse of dimensionality (intuition)

As dimensionality increases:

- data becomes sparse (needs more samples to cover space),
- distances can become less informative,
- models can overfit more easily.

Reducing dimensions can improve generalization and computational efficiency.

6.2 When should you reduce dimensions?

Common reasons:

- **Speed**: fewer features $\Rightarrow$ faster training and prediction.
- **Noise reduction**: remove low-variance directions that often contain noise.
- **Visualization**: project to 2D/3D for plots and intuition.
- **Redundancy**: features may be highly correlated; PCA compresses them.

6.3 Dimensionality reduction vs feature selection

- **Feature selection**: keep a subset of original features (easier to interpret).
- **Feature extraction**: create new features (components) from combinations of originals (e.g., PCA).

7 Principal Component Analysis (PCA)

7.1 Idea

PCA finds a new coordinate system where:

- the first axis captures the maximum variance,
- the second captures the next maximum variance while being orthogonal, etc.

Then we keep only the top k axes and drop the rest.

7.2 What is being optimized? (important intuition)

PCA can be motivated in two equivalent ways:

- **Max-variance view:** choose a unit vector v that maximizes the variance of the projection $X_c v$.
- **Min-reconstruction-error view:** choose a k-dimensional subspace that minimizes squared reconstruction error.

Both lead to the same principal components.

7.3 Steps (more detail)

Given data matrix X with n samples and d features:

1. (Often) standardize features: subtract mean and divide by std.
2. Mean-center: $X_c = X - \mu$ (where μ is the mean of each feature).
3. Compute covariance matrix:
$$C = \frac{1}{n-1} X_c^\top X_c$$
4. Compute eigenvalues λ_i and eigenvectors v_i of C.
5. Sort eigenvectors by decreasing eigenvalues.
6. Choose top-k eigenvectors and project: $Z = X_c V_k$.

7.4 Explained variance

Eigenvalue λ_i corresponds to variance captured by component i. Explained variance ratio:

$$r_i = \frac{\lambda_i}{\sum_j \lambda_j}$$

We often choose k such that $\sum_{i=1}^{k} r_i$ exceeds a threshold (e.g., 0.90), depending on the application.

7.5 How to choose k in practice

Three common approaches:

- **Cumulative EVR**: pick smallest k reaching 90%, 95%, etc.
- **Scree plot (elbow)**: choose k where additional components add little.
- **Downstream performance**: treat k as a hyperparameter and validate it (careful: do this inside CV to avoid leakage).

7.6 Why standardize?

If features have different scales (e.g., income in thousands vs age in years), PCA may focus on the largest-scale feature. Standardization makes PCA reflect correlation structure rather than raw scale.

7.7 A tiny geometric example (2D $\to$ 1D)

Suppose (after centering) the points lie roughly on the line $x_2 \approx x_1$. Then the first principal component is approximately:

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Projecting a point x to 1D gives the scalar coordinate:

$$z = x^\top v_1$$

Intuition: we keep the "diagonal direction" and drop the perpendicular direction, losing little information if points are tightly clustered around the diagonal.

7.8 PCA and SVD (brief note)

Many software libraries compute PCA using **SVD** (singular value decomposition) for numerical stability. You do not need to compute SVD by hand, but it explains why PCA works well for large matrices.

7.9 Common pitfalls

- PCA is **linear**: it cannot capture complex non-linear structure (for that, methods like t-SNE/UMAP exist).
- Components can be hard to interpret; they are combinations of features.
- Always fit PCA on **training data only** (otherwise you leak information from validation/test).

8 LDA and ICA (conceptual)

8.1 LDA

Linear Discriminant Analysis is supervised: it uses class labels and tries to find a projection that separates classes.

How it differs from PCA. PCA ignores labels and maximizes variance overall. LDA uses labels and tries to maximize between-class separation while minimizing within-class spread.

8.2 ICA

Independent Component Analysis tries to find components that are statistically independent. A classic intuition is separating mixed signals into original sources.

Where ICA is used. Blind source separation (signals), artifact removal, and some feature extraction tasks. ICA assumes the underlying sources are independent and non-Gaussian.

9 Python example (PCA)

```
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

X_scaled = StandardScaler().fit_transform(X)
pca = PCA(n_components=2)
Z = pca.fit_transform(X_scaled)
print(pca.explained_variance_ratio_)
```

10 Practice (with answers)

P1. Why is it incorrect to apply PCA on the full dataset before splitting into train/test?

Answer: PCA learns directions from data. If you fit PCA using test data, information from test influences the representation used in training, causing leakage and optimistic evaluation.

P2. PCA vs feature selection: which is easier to interpret and why?

Answer: Feature selection is easier to interpret because it keeps original features. PCA creates components that mix features, which can be harder to explain.

P3. If EVR values are $[0.60, 0.25, 0.10, 0.05]$, what is cumulative EVR for $k = 2$?

Answer: $0.60 + 0.25 = 0.85$ (85%).

11 Exit questions (with answers)

Q1. What does PCA optimize?

Answer: PCA chooses orthogonal directions that maximize variance of the projected data (sequentially), thus capturing as much variability as possible in fewer dimensions.

Q2. Why standardize before PCA?

Answer: to avoid features with larger numeric scales dominating the covariance and the principal components.

Q3. Compare PCA and LDA.

Answer: PCA is unsupervised and maximizes variance; LDA is supervised and aims to separate classes by maximizing between-class separation relative to within-class scatter.

Q4. What is explained variance?

Answer: the amount (or proportion) of total variance in the data captured by a principal component; used to choose the number of components.

12 Summary

PCA is a core dimensionality reduction tool. LDA and ICA provide alternative projections depending on whether labels are available (LDA) or independence is desired (ICA).

Sources: Mueller and Massaron, *Machine Learning for Dummies*, For Dummies, 2016.;Mitchell, *Machine Learning*, selected chapters on learning basics and evaluation.;Scikit-learn documentation, accessed 2026-02-28.